

# **Does Turnover Volatility Affect Arbitrage?**

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## **Abstract**

This paper investigates whether the volatility in a stock's turnover (TURNVOL) prevents arbitrageurs from eliminating mispricing. Mispricing is severe in high TURNVOL stocks. Among overpriced (underpriced) stocks, the high TURNVOL stocks are the most overpriced (underpriced). Additionally, overpricing (underpricing) in high TURNVOL stocks is severe during high (low) investor sentiment periods. The findings are consistent with TURNVOL limiting arbitrage. Further, the negative relationship between TURNVOL and average return, documented in prior literature, is present only in difficult-to-short stocks. TURNVOL hindering arbitrage and the asymmetry in arbitrage arising due to the difficulty in shorting stocks together explain the negative TURNVOL-return relation. The results suggest that efforts to reduce TURNVOL by regulators has the potential to improve market efficiency in emerging markets.

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## 1. Introduction

Whether or not prices reflect all available information is of interest to finance literature. If information is not immediately incorporated, the price would deviate from the fundamentals creating a profitable trading opportunity. In a frictionless world, rational arbitrageurs will trade on such opportunities and eliminate mispricing. However, in the presence of frictions, risk-averse arbitrageurs will be unable to fully exploit the mispricing in stocks with higher arbitrage risks or trading costs. Mispricing therefore can continue for a longer duration in the presence of such frictions. So far the literature has explored the effect of frictions such as transaction costs, idiosyncratic risk, and short sale constraints on arbitrage. The potential of turnover volatility in limiting arbitrage however has not received much attention. This study attempts to bridge that gap.

Turnover, the trading volume in a stock scaled by total shares outstanding, represents trading interest in a stock. Stocks with low (high) turnover tend to be less (more) liquid. The turnover in a stock varies over time and some stocks experience a sizable variation in turnover. It is possible that the volatility in turnover (TURNVOL) could affect arbitrage. Consider an arbitrageur who spots a mispricing in a stock. For a trade in the stock to be profitable, the arbitrageur must be able to initiate and exit a position without moving the price. If the position has to be held for a while, the arbitrageur would also be concerned about the potential cost to be incurred while exiting the position. There is uncertainty about the future execution cost because it is unknown at initiation. This uncertainty is higher in high TURNVOL stocks. These stocks have a higher probability of thin trading in the future which increases the cost of exiting the position. Due to this uncertainty arbitrageurs might shy away from eliminating mispricing in high TURNVOL stocks.

This paper empirically tests whether mispricing is more pronounced in high TURNVOL stocks by using Stambaugh, Yu, and Yuan (2015) mispricing scores. The mispricing score, computed as the composite score of a stock's ranking in 11 different anomalies, identifies the mispricing in a stock. Low (high) mispricing score indicates that the stock is underpriced (overpriced). As the mispricing corrects over time, the stocks that are underpriced (overpriced) in the current period earn higher (lower) risk-adjusted returns in the future periods.

In empirical tests, each month, stocks are first sorted into quintiles based on mispricing scores as of previous month. Then within each mispricing quintile, stocks are further sorted into quintiles based on TURNVOL as of previous month. TURNVOL is computed as the standard deviation of monthly turnover in the previous 60 months. The portfolio returns are value-weighted. In the underpriced (overpriced) quintile, the risk-adjusted returns increase (decrease) with TURNVOL. Also, stocks in the high TURNVOL quintile earn the highest (lowest) risk-adjusted returns among underpriced (overpriced) stocks. This implies that among the underpriced stocks, the high TURNVOL stocks were the most underpriced in the previous period. Further, the difference in returns between the underpriced and overpriced stocks increases with TURNVOL. The results are consistent with TURNVOL affecting arbitrage.

TURNVOL computed using monthly data assumes that arbitrageurs have a monthly holding horizon. If most arbitrageurs close their positions within a month, then daily turnover volatility (DTURNVOL) will be an appropriate measure. When the above analysis is repeated using DTURNVOL, computed as the standard deviation of daily turnover in the previous 3 months, the mispricing continues to be severe in high DTURNVOL stocks. In additional analysis, the stocks with higher volatility in the Amihud (2002) illiquidity measure are also prone to mispricing suggesting that the arbitrageurs are wary of the variation in liquidity

When arbitrage is hindered, investor sentiment drives the mispricing (Stambaugh, Yu, and Yuan, 2015). By using the Baker and Wurgler (2006) investor sentiment measure this paper tests whether the mispricing in high TURNVOL stocks varies with investor sentiment. Overpricing (underpricing) in high TURNVOL stocks is stronger following high (low) sentiment periods providing additional support.

Prior studies have explored how the variation in trading volume/turnover is priced in the cross section. Chordia, Subrahmanyam, and Anshuman (2001) document a negative relationship between variation in volume and average returns. The negative relationship is quite puzzling because if variation in trading is a risk then stocks with large variation should earn higher average returns. However, the negative relationship could be due to the asymmetry in arbitrage that arises due to the difficulty in correcting overpricing in the presence of short sale constraints. When short sale constraints are binding the arbitrage capital that could have been deployed to correct overpricing flows to correct underpricing in high TURNVOL stocks instead. Therefore on average high TURNVOL stocks tend to be overpriced. Supporting the argument, negative TURNVOL-average return relation is present only in difficult-to-short stocks. More support is found in individual stock Fama-Macbeth regressions. In general, negative TURNVOL stocks earn lower average returns as documented in the prior studies. However, the relationship disappears after accounting for the mispricing in high TURNVOL stocks due to limited arbitrage. The findings provide evidence that TURNVOL limiting arbitrage and arbitrage asymmetry together explain the negative TURNVOL-average return relation.

Previous literature has documented idiosyncratic volatility (IVOL) as a limiting factor to arbitrage (Pontiff, 2006). Further, it is well known that mispricing is higher in illiquid stocks. In additional analysis, the results that mispricing is severe in high TURNVOL stocks remain robust

even after controlling for IVOL and the level of liquidity proxied by Amihud (2002) illiquidity and turnover.

This paper adds to the literature by introducing TURNVOL as an additional risk faced by arbitrageurs. Mispricing is higher in high TURNVOL stocks even after controlling for other factors that limit arbitrage. Further, the mispricing in high TURNVOL stocks varies with investor sentiment. Another contribution relates to the negative relationship between TURNVOL and average return in the cross section of stocks. This paper provides a plausible arbitrage based explanation for the puzzling negative relation.

The results here have potential regulatory implications especially for the emerging markets. Efficient markets facilitate the efficient allocation of capital. The findings suggest that price efficiency can be increased if TURNVOL can be decreased. Regulators in emerging markets can improve the market efficiency by inducing a reduction in TURNVOL by taking appropriate measures. One potential measure is subsidizing market making thereby making turnover less volatile.

The remainder of the paper is organized as follows: Section 2 discusses the literature and develops the hypotheses tested in the paper, Section 3 discusses the data, Section 4 presents the results and Section 5 concludes the paper.

## **2. Literature and Hypothesis Development:**

Prior literature has investigated the factors limiting arbitrage. Shleifer and Vishny (1997) discuss how noise trader sentiment could limit arbitrage. Pontiff (2006) argues that idiosyncratic volatility is an important holding cost for arbitrageurs. Stambaugh, Yu, and Yuan (2015) combine the arbitrage risk due to idiosyncratic volatility and arbitrage asymmetry to explain the negative

idiosyncratic volatility and return relationship initially found by Ang, Hodrick, Xing, and Zhang (2006). Korajczyk, and Sadka (2004) investigate if transactions costs can explain the momentum returns reported by Jegadeesh and Titman (1993). Hong and Stein (2002), and Nagel (2005) investigate the impact of short sale constraints on returns. Diether, Malloy, Scherbina (2003) and Stambaugh, Yu, and Yuan (2012) provide support to Miller (1977) by showing that in the presence of short sale constraints stocks with higher difference of opinion have lower returns. Hou and Moskowitz (2005) investigate the frictions affecting a stock price using delayed response of price to information. However, the potential for turnover volatility to limit arbitrage has been understudied.

Uncertainty about future execution cost could make arbitrage costly. This uncertainty arises because, when initiating a trade, the arbitrageur does not know the exact cost that will be incurred when closing the position. The uncertainty is higher in high TURNVOL stocks where there is higher probability of thin trading in the future. Thin trading increases the cost of executing large trades. Due to this, the arbitrageurs might shy away from eliminating mispricing completely given this uncertainty.

If arbitrage is indeed limited in high TURNVOL stocks, then mispricing will be severe. Among underpriced stocks, high TURNVOL stocks will be more underpriced in the current period and earn higher returns in the next period. The opposite will be true for overpriced stocks. Mispricing will vary with investor sentiment in the presence of arbitrage constraints (Stambaugh, Yu, and Yuan, 2015). Overpricing is higher during high sentiment periods and underpricing is higher during low sentiment periods. The relationship between sentiment and mispricing implies that high TURNVOL stocks will be more overpriced during periods of high investor sentiment and more underpriced during periods of low investor sentiment.

Chordia, Subrahmanyam and Anshuman (2001) investigate how the variation in trading volume is priced in the cross section of stocks and find that it is negatively priced. This effect could be a result of arbitrage asymmetry arising due to the difficulty in shorting stocks. Given the difficulty in eliminating overpricing due to short sale constraints, the arbitrage capital that could not be used to correct overpricing in high TURNVOL stocks will be deployed to correct underpricing. As a result, high TURNVOL stocks will be overpriced on average and earn lower returns. The next section describes the data used to test the hypotheses.

### **3. Data**

Returns, trading volume, total shares outstanding, and stock price are from CRSP and book value is from COMPUSTAT. Stambaugh, Yu, and Yuan (2015) mispricing scores are from Yu Yuan's website, Fama and French (2015) factor returns are from Kenneth French's website and Baker and Wurgler (2006) investment sentiment series is from Jeffrey Wurgler's website. The sample period used in this paper is from January 1966 to December 2013.

Only stocks listed on NYSE/AMEX and NASDAQ are considered. NASDAQ volume is not comparable to NYSE. To make them comparable, the volume adjustment proposed by Ritter and Gao (2010) is followed.

#### *3.1 Variables*

The following variables are used in the empirical analysis in the paper.

SIZE: Market capitalization of a stock as of the previous month.

BM: Book-to-market for stocks from July of year  $t$  to June of year  $t+1$  is the book value for the fiscal reported in calendar year  $t-1$  divided by market capitalization of stock as of year end  $t-1$ . This follows Fama and French (1992). BM values are winsorized at 1% and 99% levels.

TURN: Monthly turnover of a stock as of previous month. Turnover is defined as the trading volume in a stock divided by total shares outstanding.

TURNVOL: Standard deviation of monthly turnover computed using the previous 60 months of turnover. A stock should have at least 18 months of turnover data in the previous 60 months.

DTURNVOL: Standard deviation of daily turnover computed using previous 3 months of daily turnover. A stock should have at least 18 days of daily turnover data in the previous 3 months.

AMIHUD: Amihud (2002) illiquidity measure as of the previous month computed using daily return and volume data in the month. AMIHUD illiquidity for the month  $t$  for stock  $i$  is calculated as

$$AMIHUD_{it} = \frac{1}{T} \sum_{d=1}^T \frac{|R_{id}|}{DVOL_{id}}$$

where  $|R_{id}|$  is the absolute return of the stock  $i$  on day  $d$  of the month  $t$ . DVOL is the dollar volume in the stock for that day.

AMIHUDVOL: Volatility in AMIHUD illiquidity measure computed using the previous 60 months of data. A stock should have a minimum of 18 months of AMIHUD illiquidity data in the previous 60 months.

1/PRICE: Reciprocal of the price of a stock as of previous month.

IVOL: Standard deviation of residuals obtained by regressing daily returns each month on Fama and French 3 factors. This methodology follows Stambaugh, Yu, and Yuan (2015). IVOL is computed only for stocks with at least 18 return observations in a month.



MISPRICING: Stambaugh, Yu and Yuan (2015) construct a measure of mispricing based on a stock's composite ranking in the following 11 anomalies.

- (a) Net stock issues
- (b) Composite equity issues
- (c) Accruals
- (d) Net Operating Assets
- (e) Asset Growth
- (f) Investment-to-Assets
- (g) Distress
- (h) O-score
- (i) Momentum
- (j) Gross Profitability Premium
- (k) Return on Assets

RET23: For the month  $t$ , RET23 is the cumulative return in the months  $t-2$  and  $t-3$ .

RET46: For the month  $t$ , RET46 is the cumulative return in the months from  $t-4$  to  $t-6$ .

RET712: RET712 is the cumulative return in the months from  $t-7$  to  $t-12$ .

SENTIMENT: Baker and Wurgler (2006) sentiment measure is the first principal component of the following six measures

- (a) Closed-end fund discount
- (b) No of IPOs
- (c) IPO first-day returns
- (d) NYSE Share Turnover
- (e) Equity share in total new issues
- (f) Dividend premium

## 4. Results

Each month, stocks are sorted into quintiles on the mispricing score as of the previous month. The stocks in the quintile with lowest mispricing score are the most underpriced stocks and stocks in the highest mispricing score quintile are the most overpriced stocks. If TURNVOL limits arbitrage, then among underpriced stocks the high TURNVOL stocks will be most underpriced and earn higher returns than low TURNVOL stocks. To test this, within each mispricing quintile stocks are in turn sorted into quintiles on TURNVOL. Value weighted risk-adjusted returns are computed using the Fama and French (2015) five factor model with investment and profitability as new factors in addition to market, size and value factors.

Table 1, Panel A presents the average market capitalization of the stocks in each group. Underpriced stocks are relatively larger and overpriced stocks are relatively smaller in size. This is due to the difficulty in shorting small stocks (D'avolio, 2002). Within each mispricing quintile, the high TURNVOL stocks are smaller in size than the low TURNVOL stocks since large stocks have relatively stable turnover compared to small stocks.

Table 1, Panel B presents the average standard deviation of monthly turnover in each group. Turnover volatility is high in overpriced stocks compared to underpriced stocks. In overpriced stocks, short sale constraints prevent pessimistic investors from trading. When price is below fundamentals investors will be less pessimistic and trading will not be affected as much. However, when the stock price is above fundamentals trading short sale constraints will be binding resulting in thin trading. Hence the stocks with a tendency to be overpriced have higher TURNVOL.

#### *4.1 Turnover Volatility and Mispricing*

Table 3 presents the risk adjusted returns of the value weighted portfolios formed by sorting first on mispricing and then on TURNVOL. The first row reports the risk-adjusted returns of the stocks in the most underpriced group. Underpriced stocks earn higher returns the next period as the mispricing is corrected. The returns increase with TURNVOL suggesting that underpricing increases with TURNVOL. The high TURNVOL stocks earn the highest returns consistent with them being most underpriced. The last column reports the risk-adjusted returns of long-short portfolios formed by buying high TURNVOL stocks and shorting low TURNVOL stocks within each mispricing group. The difference is 60 basis points a month and is statistically significant.

Among the most overpriced stocks, the risk-adjusted returns decrease with TURNVOL. The high TURNVOL stocks, since they are most overpriced earn the lowest returns next period. In the last column, the long-short TURNVOL portfolio alpha is negative and statistically significant. The last two rows report the difference between alphas and respective t-statistics of the most underpriced stocks and the most overpriced stocks within each TURNVOL quintile. This is a measure of mispricing. The magnitude of mispricing increases with TURNVOL. The results are consistent with mispricing being severe in high TURNVOL stocks.

#### *4.2 Other measures of turnover volatility*

Previous tests used monthly turnover volatility measure following Chordia, Subrahmanyam and Anshuman (2001) who use monthly variation in trading volume to test whether variation in trading volume is priced in the cross-section of returns. In limits to arbitrage tests, it is important that the period used to compute TURNVOL is comparable to the arbitrageurs holding period. Active mutual funds turnover stocks about once a year. But some hedge funds can

flip holdings faster. Table 4, reports the risk-adjusted returns of portfolios sorted on mispricing and a measure of turnover volatility computed from daily turnover (DTURNVOL). Results are qualitatively similar to previous tests. Among underpriced stocks the Fama and French 5-factor alpha increases with DTURNVOL and among overpriced stocks the alpha decreases with turnover but is not monotonic. Long-short DTURNVOL portfolio has positive and significant alpha in underpriced stocks and negative alpha in overpriced stocks significant at the 10% level. The findings provide evidence that results are not sensitive to the period used to compute turnover volatility.

The results so far suggest that the arbitrageurs might care about variation in liquidity. This is tested by repeating the previous analysis by using the variation in Amihud (2002) illiquidity measure (AMIHUDVOL) instead of TURNVOL. From Table 2, TURNVOL and AMIHUDVOL are not highly correlated. As liquidity is multi-dimensional both measures could be capturing different aspects of variation in liquidity. From Table 5, among underpriced stocks the alpha increases with AMIHUDVOL and among overpriced stocks the alpha decreases with AMIHUDVOL. The long-short AMIHUDVOL portfolio has positive and significant alpha in underpriced stocks and negative alpha in overpriced stocks significant at the 10% level. From the last two rows, we find that the magnitude of mispricing increases monotonically with AMIHUDVOL. The results are consistent with variation in liquidity affecting arbitrage.

#### *4.3 Sentiment and Mispricing*

This section investigates how the relationship between TURNVOL and mispricing is affected by investor sentiment. Baker and Wurgler (2006) sentiment (BW) is the measure of investor sentiment used. In the presence of arbitrage costs, as arbitrageurs are unable to eliminate mispricing, sentiment will drive the mispricing (Stambaugh, Yu and Yuan, 2015). When sentiment

is high overpricing will be larger. Because high TURNVOL stocks is where arbitrageurs will be hindered the most, we should see high TURNVOL stocks in the most overpriced quintile having lower returns following high sentiment periods. Similarly the most underpriced stocks with high TURNVOL should earn higher returns following low sentiment periods.

Table 6 reports the risk-adjusted alpha for high and low sentiment months. Months are classified as high and low sentiment depending on whether BW sentiment measure was higher or lower than median respectively. Following high sentiment months high TURNVOL stocks in the most overpriced quintile earn lower returns. Following low sentiment months, high TURNVOL stocks in the most underpriced quintile earn higher returns. This provides additional support that TURNVOL hinders arbitrage and hence sentiment drives mispricing in high TURNVOL stocks.

#### *4.4 Arbitrage Asymmetry and negative TURNVOL-return relation*

Arbitrage asymmetry is the difficulty in correcting overpricing due to short-sale constraints. It is easier to correct underpricing as there are no constraints on the long side. Stambaugh, Yu, and Yuan (2015) note that due to the arbitrage asymmetry, more arbitrage capital will be deployed to correct underpricing. As a result overpricing will continue to exist in stocks with short-sale constraints.

Chordia, Subrahmanyam, and Anshuman (2001) document that stocks with high variation in volume earn low returns. The relation could arise due to arbitrage asymmetry. If so the relation will be stronger in difficult-to-short stocks.

Institutional ownership (IO) serves as a measure of short sale constraints (Nagel, 2005). Stocks are first sorted into quintiles based on IO as of the previous month. Then within each IO quintile they are in turn sorted into quintiles based on TURNVOL. Table 7 presents the results. In

the lowest IO group, the group with highest short sale constraints, stocks with high TURNVOL earn negative risk-adjusted returns. In TURNVOL quintiles 3 and 4 the negative risk-adjusted return are significant. This is not monotonic since the highest TURNVOL quintile in the lowest IO group earns less negative return compared to group 4. The high TURNVOL stocks in other IO groups do not show the negative-return relation. In the last row and last column of the table, the difference in returns between long-short portfolio IO portfolios formed by buying high TURNVOL stocks and selling low TURNVOL stocks is negative and significant at 10% level supporting the arbitrage asymmetry claim. Next section provides additional support to this using Fama-Macbeth analysis.

#### *4.5 Individual stocks Fama-Macbeth analysis*

Table 8 reports the results of the Fama-Macbeth regression of individual stock returns on characteristics. Individual stocks risk adjusted returns are computed following Brennan, Chordia and Subrahmanyam (1998). The characteristics considered are SIZE, BM, 1/PRICE, RET23, RET46, RET712, Mispricing and Mispricing interacted with TURNVOL. Natural logarithm of all variables is used to control for skewness with the exception of mispricing and other return based variables. MISPRICING is a continuous variable with high value suggesting overpricing.

In the first column of Table 8 TURNVOL has a negative coefficient. This is consistent with the findings in Chordia, Subrahmanyam, and Anshuman (2001). This is the negative TURNVOL-return relation puzzle. High TURNVOL stocks earn lower risk adjusted returns. In the second column, mispricing and the interaction of mispricing and TURNVOL are added. The coefficient on TURNVOL now becomes insignificant. The coefficient on the interaction term is negative and significant. This suggests that it is the high TURNVOL stocks that are overpriced that earn negative returns. This provides additional support to the arbitrage asymmetry claim

discussed in the previous section. TURNVOL as a limiting factor to arbitrage and arbitrage asymmetry together explain the negative TURNVOL-average return relation.

#### *4.6 Effect of TURNVOL on arbitrage after controlling for other limiting factors to arbitrage*

For turnover volatility to be an additional factor limiting arbitrage it must explain the mispricing after controlling for previously documented limiting factors. To test this, each month stocks are sorted into three groups on mispricing scores. Then within each mispricing tercile, stocks are sorted into three groups on Amihud (2002) illiquidity measure(AMIHUD). Then within each mispricing-AMIHUD group, stocks are sorted into three groups based on TURNVOL.

Table 9 reports the risk-adjusted returns of the value weighted portfolios. In the most underpriced group, across all the AMIHUD terciles, alphas increase with TURNVOL. We find that the long-short TURNVOL alphas are higher among very liquid stocks with lower levels of AMIHUD illiquidity. The last row provides the alpha of TURNVOL terciles after controlling for illiquidity. The long-short TURNVOL portfolio alpha across all stocks after accounting for AMIHUD is positive and significant 65 basis points per month.

In the most overpriced group, we find that alpha decreases with TURNVOL across all AMIHUD terciles with the decrease being strongest among the most illiquid stocks.

Table 10 repeats the analysis using Turnover (TURN) as a measure of liquidity. In the most underpriced group, across all the TURN terciles, alphas increase with TURNVOL. The long-short TURNVOL alphas are higher among very liquid stocks with high turnover. The last row provides the alpha of TURNVOL terciles after controlling for level of TURN. The long-short TURNVOL portfolio alpha across all stocks after controlling for TURN is positive and significant 54 basis points per month.

In the most overpriced group, the alpha decreases with TURNVOL across all TURN terciles with the decrease being strongest in the low TURN stocks. Across all stocks, in the overpriced group, after controlling for TURN, alpha decreases with TURNVOL and the long-short TURNVOL portfolio returns are statistically significant. The results suggest that TURNVOL makes arbitrage costly over and above liquidity.

Pontiff (2006) argues that idiosyncratic volatility (IVOL) is an important holding cost incurred by the arbitrageurs. In Table 11, the previous 3x3x3 analysis is repeated by replacing measures of liquidity with IVOL. In the most underpriced group, the alphas increase with TURNVOL across all IVOL groups. In the most overpriced group, the alphas decrease with TURNVOL across all IVOL groups. Across all stocks, after controlling for IVOL, the long-short TURNVOL portfolios earn positive and significant returns in the underpriced group and negative and significant returns in the overpriced group respectively. The evidence provided in this section suggests that TURNVOL limits arbitrage over and above previously documented limits to arbitrage.

## **5. Conclusion:**

This paper highlights an important holding cost faced by the arbitrageurs: the uncertainty in future turnover. The uncertainty arises because a stock's turnover in the future is unknown to the arbitrageur while initiating a position. Arbitrageurs will not eliminate mispricing given the uncertainty. Consistent with the claim mispricing is severe in high TURNVOL stocks. Among overpriced stocks, high TURNVOL stocks are more overpriced and earn lower returns subsequently. In high TURNVOL stocks, overpricing is severe during periods of high investor sentiment and underpricing is severe during periods of low sentiment. The relationship holds after



controlling for level of liquidity, idiosyncratic volatility, different risk adjustments and other characteristics.

Prior literature has documented a negative relationship between variation in a stock's trading volume and average returns in the cross section. The negative TURNVOL – return relationship is found only in stocks with short sale constraints. This is the result of asymmetry in arbitrage arising due to the difficulty in eliminating overpricing in the presence of short sale constraints. Therefore, more arbitrage capital flows to correct underpricing. Consequently, the high TURNVOL stocks are overpriced on average and therefore earn low returns. This provides an arbitrage based explanation for the relationship.

The results here have potential regulatory implications especially for the emerging markets. Efficient markets facilitate the efficient allocation of capital. The findings suggest that price efficiency can be increased if TURNVOL can be decreased. Regulators in emerging markets can improve the market efficiency by inducing a reduction in TURNVOL by taking appropriate measures. One potential measure is subsidizing market making thereby making turnover less volatile.

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**Table 1: Average Size and Turnover Volatility of Stocks sorted on Mispricing and Turnover Volatility**

The table presents the average market capitalization (in \$ Millions) and standard deviation of monthly turnover (TURNVOL) of stocks in the mispricing and turnover volatility quintiles. At the beginning of each month, stocks are sorted into quintiles based on their mispricing scores as of the previous month. Within each mispricing quintile, the stocks are in turn sorted into quintiles on the standard deviation of monthly turnover as of the previous month. Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. Standard deviation is computed using previous 60 months of turnover data. Sample period is from January 1966 to December 2013.

|                                                  | Lowest TURNVOL | 2     | 3     | 4     | Highest TURNVOL |
|--------------------------------------------------|----------------|-------|-------|-------|-----------------|
| Panel A : Market Cap (in \$ Millions)            |                |       |       |       |                 |
| Most Underpriced                                 | 7,705          | 5,614 | 3,260 | 1,909 | 1,495           |
| 2                                                | 3,304          | 3,376 | 2,403 | 1,813 | 1,680           |
| 3                                                | 1,529          | 2,336 | 1,824 | 1,534 | 1,358           |
| 4                                                | 1,254          | 1,755 | 1,444 | 1,174 | 1,100           |
| Most Overpriced                                  | 1,002          | 1,371 | 1,002 | 908   | 942             |
| Panel B : Standard Deviation of Monthly Turnover |                |       |       |       |                 |
| Most Underpriced                                 | 0.010          | 0.020 | 0.030 | 0.047 | 0.107           |
| 2                                                | 0.009          | 0.020 | 0.031 | 0.048 | 0.111           |
| 3                                                | 0.008          | 0.019 | 0.031 | 0.050 | 0.120           |
| 4                                                | 0.008          | 0.020 | 0.034 | 0.054 | 0.132           |
| Most Overpriced                                  | 0.011          | 0.026 | 0.043 | 0.067 | 0.159           |

**Table 2: Correlations**

The table reports the correlation between the variables used in the paper. TURNVOL is the standard deviation of monthly turnover computed using the turnover from previous 60 months. DTURNVOL is the standard deviation of daily turnover computed using previous 3 months of daily data. AMIHUD is the monthly Amihud (2002) illiquidity measure. AMIHUDVOL is the volatility in AMIHUD illiquidity measure computed using the monthly AMIHUD measure from previous 60 months. IVOL is the standard deviation of return residuals from Fama and French 3 factor model computed using daily returns in the previous month. TURN is the ratio of trading volume in the previous month and total shares outstanding. SIZE is the market capitalization of the stock as of the previous month. Reported numbers are cross sectional averages of individual stock correlations. Sample period is from January 1966 to December 2013.

|           | TURNVOL | DTURNVOL | AMIHUDVOL | IVOL  | AMIHUD | TURN  | SIZE  |
|-----------|---------|----------|-----------|-------|--------|-------|-------|
| TURNVOL   | 1.00    | 0.35     | -0.13     | 0.03  | -0.14  | 0.33  | 0.19  |
| DTURNVOL  | 0.35    | 1.00     | -0.05     | 0.12  | -0.19  | 0.65  | 0.19  |
| AMIHUDVOL | -0.13   | -0.05    | 1.00      | 0.14  | 0.34   | -0.10 | -0.28 |
| IVOL      | 0.03    | 0.12     | 0.14      | 1.00  | 0.29   | 0.26  | -0.19 |
| AMIHUD    | -0.14   | -0.19    | 0.34      | 0.29  | 1.00   | -0.27 | -0.39 |
| TURN      | 0.33    | 0.65     | -0.10     | 0.26  | -0.27  | 1.00  | 0.25  |
| SIZE      | 0.19    | 0.19     | -0.28     | -0.19 | -0.39  | 0.25  | 1.00  |

**Table 3: Risk-Adjusted Returns of Portfolios sorted on Mispricing and TURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing and TURNVOL. At the beginning of each month, stocks are sorted into quintiles based on their mispricing scores as of the previous month. Within each mispricing quintile, the stocks are in turn sorted into quintiles on the standard deviation of *monthly turnover* as of the previous month (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. TURNVOL is computed from 60 months of prior monthly turnover data. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                                       | Lowest<br>TURNVOL | 2                 | 3                 | 4                 | Highest<br>TURNVOL | Highest - Lowest<br>TURNVOL |
|---------------------------------------|-------------------|-------------------|-------------------|-------------------|--------------------|-----------------------------|
| Most Underpriced                      | -0.05%<br>(-0.64) | -0.02%<br>(-0.26) | 0.24%<br>(2.60)   | 0.46%<br>(4.02)   | 0.55%<br>(3.49)    | 0.60%<br>(3.39)             |
| 2                                     | -0.08%<br>(-0.90) | -0.13%<br>(-1.61) | -0.04%<br>(-0.56) | 0.21%<br>(1.96)   | 0.49%<br>(3.16)    | 0.57%<br>(3.23)             |
| 3                                     | -0.20%<br>(-1.74) | 0.00%<br>(0.02)   | -0.16%<br>(-1.79) | -0.04%<br>(-0.36) | 0.33%<br>(2.23)    | 0.52%<br>(2.55)             |
| 4                                     | -0.15%<br>(-1.28) | -0.31%<br>(-2.85) | -0.07%<br>(-0.76) | -0.29%<br>(-2.43) | 0.06%<br>(0.39)    | 0.21%<br>(1.03)             |
| Most Overpriced                       | -0.25%<br>(-2.03) | -0.19%<br>(-1.37) | -0.51%<br>(-3.93) | -0.79%<br>(-5.73) | -0.85%<br>(-5.00)  | -0.60%<br>(-3.05)           |
| Most Underpriced -<br>Most Overpriced | 0.20%<br>(1.27)   | 0.17%<br>(0.92)   | 0.75%<br>(4.29)   | 1.25%<br>(6.59)   | 1.40%<br>(6.19)    | 1.20%<br>(4.92)             |

**Table 4: Risk-Adjusted Returns of Portfolios sorted on Mispricing and DTURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing and DTURNVOL. At the beginning of each month, stocks are sorted into quintiles based on their mispricing scores as of the previous month. Within each mispricing quintile, the stocks are in turn sorted into quintiles on the standard deviation of daily turnover as of the previous month(DTURNVOL). Daily Turnover is defined as the ratio of daily trading volume and total shares outstanding in a stock. DTURNVOL is computed from 3 months of prior daily turnover data. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                                       | Lowest<br>DTURNVOL | 2                 | 3                 | 4                 | Highest<br>DTURNVOL | Highest - Lowest<br>DTURNVOL |
|---------------------------------------|--------------------|-------------------|-------------------|-------------------|---------------------|------------------------------|
| Most Underpriced                      | -0.07%<br>(-0.83)  | 0.02%<br>(0.27)   | 0.27%<br>(2.86)   | 0.33%<br>(3.21)   | 0.63%<br>(4.23)     | 0.69%<br>(3.90)              |
| 2                                     | -0.01%<br>(-0.13)  | -0.06%<br>(-0.74) | -0.02%<br>(-0.21) | 0.13%<br>(1.41)   | 0.22%<br>(1.44)     | 0.23%<br>(1.36)              |
| 3                                     | -0.30%<br>(-2.64)  | -0.11%<br>(-1.29) | -0.03%<br>(-0.31) | 0.06%<br>(0.60)   | 0.22%<br>(1.66)     | 0.52%<br>(2.84)              |
| 4                                     | -0.20%<br>(-1.73)  | -0.19%<br>(-1.97) | -0.19%<br>(-2.04) | -0.04%<br>(-0.42) | -0.20%<br>(-1.51)   | 0.00%<br>(-0.02)             |
| Most Overpriced                       | -0.40%<br>(-3.30)  | -0.39%<br>(-3.00) | -0.53%<br>(-4.28) | -0.40%<br>(-3.04) | -0.77%<br>(-4.76)   | -0.37%<br>(-1.90)            |
| Most Underpriced -<br>Most Overpriced | 0.33%<br>(2.22)    | 0.42%<br>(2.37)   | 0.80%<br>(4.54)   | 0.73%<br>(4.17)   | 1.40%<br>(6.63)     | 1.06%<br>(4.44)              |

**Table 5: Risk-Adjusted Returns of Portfolios sorted on Mispricing and AMIHUDVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing and AMIHUDVOL. At the beginning of each month, stocks are sorted into quintiles based on their mispricing scores as of the previous month. Within each mispricing quintile, the stocks are in turn sorted into quintiles on the standard deviation of monthly Amihud(2002) Illiquidity measure(AMIHUDVOL). AMIHUDVOL is computed from previous 60 months of AMIHUD measure. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                                       | Lowest<br>AMIHUDVOL | 2                 | 3                 | 4                 | Highest<br>AMIHUDVOL | Highest - Lowest<br>AMIHUDVOL |
|---------------------------------------|---------------------|-------------------|-------------------|-------------------|----------------------|-------------------------------|
| Most Underpriced                      | 0.09%<br>(1.74)     | 0.19%<br>(2.53)   | 0.34%<br>(4.33)   | 0.34%<br>(3.40)   | 0.60%<br>(5.50)      | 0.51%<br>(4.32)               |
| 2                                     | 0.01%<br>(0.18)     | 0.11%<br>(1.32)   | 0.21%<br>(2.37)   | 0.27%<br>(2.90)   | 0.32%<br>(2.47)      | 0.31%<br>(2.25)               |
| 3                                     | -0.05%<br>(-0.92)   | 0.26%<br>(2.14)   | 0.28%<br>(2.78)   | 0.24%<br>(2.38)   | 0.07%<br>(0.59)      | 0.12%<br>(0.90)               |
| 4                                     | -0.21%<br>(-2.95)   | 0.07%<br>(0.66)   | 0.06%<br>(0.65)   | -0.15%<br>(-1.47) | 0.13%<br>(0.85)      | 0.33%<br>(1.86)               |
| Most Overpriced                       | -0.42%<br>(-3.63)   | -0.38%<br>(-3.06) | -0.49%<br>(-5.41) | -0.66%<br>(-5.98) | -0.76%<br>(-5.23)    | -0.33%<br>(-1.88)             |
| Most Underpriced<br>- Most Overpriced | 0.51%<br>(3.38)     | 0.58%<br>(3.60)   | 0.82%<br>(6.56)   | 1.00%<br>(6.37)   | 1.36%<br>(8.11)      | 0.86%<br>(4.01)               |



**Table 6: Risk Adjusted Returns of portfolios sorted on Mispricing and TURNVOL in High-Sentiment and Low-Sentiment Periods.**

The table presents the Fama French three factor alpha of portfolios ranked on mispricing and TURNVOL for High Sentiment and Low Sentiment months. Each month, stocks are sorted into quintiles based on their mispricing scores as of previous month. Within each mispricing quintile, they are sorted in turn into quintiles on the standard deviation of monthly turnover (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock and is computed using 60 months of previous turnover data. Sample period is from January 1966 to December 2013. Reported numbers are  $a_H$  and  $a_L$  in the regression below.  $d_{H,t}$  is a dummy variable that takes value of 1 if the Baker and Wurgler(2006) investment sentiment measure was above median previous month and  $d_{L,t}$  is the dummy variable that takes value of 1 if the sentiment in the previous month was below median . Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

$$R_{i,t} = a_H d_{H,t} + a_L d_{L,t} + b MKT_t + c SMB_t + d HML_t + e CMA_t + f RMW_t + \epsilon_{i,t}$$

|                                       | High Sentiment months |                    |                     | Low Sentiment months |                    |                     |
|---------------------------------------|-----------------------|--------------------|---------------------|----------------------|--------------------|---------------------|
|                                       | Lowest<br>TURNVOL     | Highest<br>TURNVOL | Highest -<br>Lowest | Lowest<br>TURNVOL    | Highest<br>TURNVOL | Highest -<br>Lowest |
| Most Underpriced                      | 0.13%<br>(0.84)       | 0.24%<br>(0.89)    | 0.10%<br>(0.32)     | -0.12%<br>(-1.23)    | 0.43%<br>(2.09)    | 0.55%<br>(2.39)     |
| 2                                     | -0.09%<br>(-0.59)     | -0.08%<br>(-0.29)  | 0.02%<br>(0.06)     | -0.04%<br>(-0.33)    | 0.54%<br>(2.44)    | 0.57%<br>(2.32)     |
| 3                                     | -0.28%<br>(-1.45)     | -0.18%<br>(-0.72)  | 0.10%<br>(0.29)     | -0.06%<br>(-0.38)    | 0.41%<br>(2.09)    | 0.47%<br>(1.69)     |
| 4                                     | 0.19%<br>(0.85)       | -0.07%<br>(-0.24)  | -0.26%<br>(-0.66)   | -0.23%<br>(-1.44)    | 0.09%<br>(0.44)    | 0.32%<br>(1.15)     |
| Most Overpriced                       | 0.28%<br>(1.26)       | -0.55%<br>(-1.79)  | -0.82%<br>(-2.21)   | -0.37%<br>(-2.34)    | -0.58%<br>(-2.61)  | -0.21%<br>(-0.77)   |
| Most Underpriced - Most<br>Overpriced | -0.14%<br>(-0.50)     | 0.78%<br>(1.91)    | 0.93%<br>(2.00)     | 0.26%<br>(1.30)      | 1.01%<br>(3.59)    | 0.76%<br>(2.39)     |

**Table 7: Risk-Adjusted Returns of Portfolios sorted on Institutional Ownership and TURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on Institutional Ownership (IO) and TURNVOL. At the beginning of each month, stocks are sorted into quintiles based on institutional ownership as of the previous month. Within each short interest quintile, the stocks are in turn sorted into quintiles on the standard deviation of *monthly turnover* as of the previous month (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. TURNVOL is computed from 60 months of prior monthly turnover data. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                           | Lowest<br>TURNVOL | 2                 | 3                 | 4                 | Highest<br>TURNVOL | Highest - Lowest<br>TURNVOL |
|---------------------------|-------------------|-------------------|-------------------|-------------------|--------------------|-----------------------------|
| Lowest IO                 | -0.06%<br>(-0.28) | 0.45%<br>(1.45)   | -0.48%<br>(-1.97) | -0.61%<br>(-2.77) | -0.36%<br>(-1.10)  | -0.30%<br>(-0.78)           |
| 2                         | 0.08%<br>(0.44)   | 0.02%<br>(0.13)   | -0.12%<br>(-0.61) | 0.04%<br>(0.19)   | 0.35%<br>(1.18)    | 0.27%<br>(0.73)             |
| 3                         | 0.14%<br>(0.82)   | 0.30%<br>(1.39)   | 0.18%<br>(0.92)   | 0.22%<br>(1.18)   | 0.39%<br>(1.48)    | 0.25%<br>(0.82)             |
| 4                         | -0.15%<br>(-1.76) | -0.04%<br>(-0.32) | 0.46%<br>(3.30)   | 0.15%<br>(0.95)   | 0.37%<br>(1.85)    | 0.52%<br>(2.24)             |
| Highest IO                | -0.30%<br>(-2.83) | -0.31%<br>(-2.88) | -0.10%<br>(-0.93) | 0.21%<br>(1.50)   | 0.12%<br>(0.78)    | 0.42%<br>(2.19)             |
| Lowest IO –<br>Highest IO | 0.24%<br>(1.12)   | 0.76%<br>(2.34)   | -0.39%<br>(-1.39) | -0.82%<br>(-2.98) | -0.48%<br>(-1.35)  | -0.72%<br>(-1.79)           |

**Table 8: Fama-Macbeth Regression of Individual Risk Adjusted Returns on Characteristics**

The table reports the Fama Macbeth Regression coefficients of individual risk adjusted stock return on Characteristics using the methodology in Brennan, Chordia and Subrahmanyam (1998). Individual stock excess return is risk adjusted using Fama- French five factors. Factor loadings are allowed to vary over time and are computed from previous 60 months of returns. Natural logarithm of all variables is used with the exception of mispricing, RET23, RET46, RET712. SIZE refers to market capitalization, BM refers to the book to market, 1/PRICE is the reciprocal of price, and TURNVOL is the standard deviation of turnover as of previous month computed from monthly turnover data in the prior 60 months. Mispricing is the Stambaugh, Yu, and Yuan(2015) mispricing score. RET23 refers to the return in the second and third month previous to current month. RET46 is the buy and hold return of the stocks from six month to four months before the current month. RET712 refers to the buy and hold return of the stock from twelve month to seven month before the current month. Sample period is from Jan 1966 to Dec 2013. Fama-Macbeth t-statistics in parenthesis.

| Variable             | Coefficient       | Coefficient       |
|----------------------|-------------------|-------------------|
| Constant             | 1.420<br>(3.66)   | 3.724<br>(9.26)   |
| SIZE                 | -0.060<br>(-2.83) | -0.053<br>(-2.62) |
| BM                   | 0.162<br>(4.46)   | 0.091<br>(2.58)   |
| 1/PRICE              | 0.086<br>(1.25)   | 0.199<br>(3.36)   |
| RET23                | 0.397<br>(1.62)   | 0.210<br>(0.82)   |
| RET46                | 0.607<br>(3.08)   | 0.268<br>(1.36)   |
| RET712               | 0.287<br>(1.94)   | 0.287<br>(1.94)   |
| TURNVOL              | -0.151<br>(-4.48) | 0.107<br>(1.55)   |
| Mispricing           |                   | -0.038<br>(-7.03) |
| Mispricing * TURNVOL |                   | -0.004<br>(-3.15) |

**Table 9: Risk-Adjusted Returns of Portfolios sorted on Mispricing , AMIHUD and TURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing, AMIHUD and TURNVOL. At the beginning of each month, stocks are sorted into three groups based on their mispricing scores as of the previous month. Within each mispricing groups, the stocks are in turn sorted into terciles on the monthly Amihud (2002) measure(AMIHUD) as of the previous month. Within each mispricing and AMIHUD groups, the stocks are in turn sorted into terciles on the standard deviation of *monthly turnover* as of the previous month (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. TURNVOL is computed from 60 months of prior monthly turnover data. In the table TURNVOL is reported as TVOL. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                      | Most Underpriced  |                 |                   |                   | 2                 |                   | Most Overpriced   |                   |                   |                   |                   |                   |
|----------------------|-------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                      | Low<br>TVOL       | 2               | High<br>TVOL      | High -Low<br>TVOL | Low<br>TVOL       | 2                 | High<br>TVOL      | High -Low<br>TVOL | Low<br>TVOL       | 2                 | High<br>TVOL      | High -Low<br>TVOL |
| Low AMIHUD           | -0.12%<br>(-2.31) | 0.28%<br>(3.93) | 0.53%<br>(4.48)   | 0.65%<br>(4.82)   | -0.14%<br>(-2.07) | 0.00%<br>(0.06)   | 0.35%<br>(3.15)   | 0.13%<br>(3.35)   | -0.07%<br>(-0.76) | -0.48%<br>(-4.64) | -0.41%<br>(-3.06) | -0.34%<br>(-2.15) |
| 2                    | -0.04%<br>(-0.64) | 0.17%<br>(2.62) | 0.48%<br>(5.51)   | 0.53%<br>(4.95)   | -0.12%<br>(-1.68) | -0.01%<br>(0.10)  | 0.04%<br>(0.44)   | 0.16%<br>(1.37)   | -0.28%<br>(-2.97) | -0.54%<br>(-5.77) | -0.67%<br>(-5.41) | -0.39%<br>(-2.65) |
| High AMIHUD          | 0.15%<br>(1.75)   | 0.38%<br>(4.67) | 0.28%<br>(2.79)   | 0.12%<br>(1.01)   | -0.04%<br>(-0.41) | -0.16%<br>(-1.72) | -0.29%<br>(-2.71) | -0.25%<br>(-1.95) | -0.50%<br>(-4.06) | -0.84%<br>(-7.59) | -1.12%<br>(-8.58) | -0.63%<br>(-4.33) |
| High - Low<br>AMIHUD | 0.27%<br>(2.75)   | 0.10%<br>(0.94) | -0.26%<br>(-1.56) |                   | 0.10%<br>(0.88)   | -0.16%<br>(-1.28) | -0.64%<br>(-4.09) |                   | -0.42%<br>(-3.39) | -0.37%<br>(-2.60) | -0.71%<br>(-4.10) |                   |
| All Stocks           | -0.11%<br>(-2.30) | 0.28%<br>(4.30) | 0.54%<br>(4.97)   | 0.65%<br>(5.22)   | -0.14%<br>(-2.18) | 0.00%<br>(0.06)   | 0.31%<br>(3.09)   | 0.44%<br>(3.32)   | -0.11%<br>(-1.17) | -0.50%<br>(-5.32) | -0.47%<br>(-3.88) | -0.37%<br>(-2.58) |

**Table 10: Risk-Adjusted Returns of Portfolios sorted on Mispricing , TURN and TURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing, TURN and TURNVOL. At the beginning of each month, stocks are sorted into three groups based on their mispricing scores as of the previous month. Within each mispricing groups, the stocks are in turn sorted into terciles on the monthly turnover (TURN) as of the previous month. Within each mispricing and TURN groups, the stocks are in turn sorted into terciles on the standard deviation of *monthly turnover* as of the previous month (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. TURNVOL is computed from 60 months of prior monthly turnover data. In the table TURNVOL is reported as TVOL. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                    | Most Underpriced  |                   |                 |                  |                   | Most Overpriced   |                   |                   |                   |                   |                   |                   |
|--------------------|-------------------|-------------------|-----------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                    | Low<br>TVOL       | 2                 | High<br>TVOL    | High-Low<br>TVOL | Low<br>TVOL       | 2                 | High<br>TVOL      | High-Low<br>TVOL  | Low<br>TVOL       | 2                 | High<br>TVOL      | High-Low<br>TVOL  |
| Low TURN           | -0.17%<br>(-1.72) | -0.26%<br>(-2.64) | 0.01%<br>(0.11) | 0.18%<br>(1.35)  | -0.13%<br>(-1.10) | -0.14%<br>(-1.37) | -0.24%<br>(-2.10) | -0.11%<br>(-0.68) | -0.21%<br>(-1.68) | -0.52%<br>(-4.40) | -0.97%<br>(-7.57) | -0.76%<br>(-4.73) |
| 2                  | -0.07%<br>(-0.94) | 0.21%<br>(2.28)   | 0.23%<br>(2.14) | 0.30%<br>(2.38)  | -0.13%<br>(-1.56) | -0.25%<br>(-2.71) | -0.13%<br>(-1.11) | 0.00%<br>(-0.02)  | -0.10%<br>(-0.85) | -0.33%<br>(-2.62) | -0.63%<br>(-4.15) | -0.53%<br>(-3.27) |
| High TURN          | 0.16%<br>(1.81)   | 0.29%<br>(2.38)   | 0.66%<br>(3.94) | 0.50%<br>(3.12)  | -0.07%<br>(-0.80) | 0.20%<br>(1.64)   | 0.28%<br>(1.67)   | 0.35%<br>(1.87)   | -0.35%<br>(-2.94) | -0.55%<br>(-4.14) | -0.60%<br>(-3.40) | -0.25%<br>(-1.49) |
| High - Low<br>TURN | 0.33%<br>(2.28)   | 0.56%<br>(3.40)   | 0.65%<br>(3.24) |                  | 0.06%<br>(0.37)   | 0.34%<br>(1.94)   | 0.51%<br>(2.42)   |                   | -0.14%<br>(-0.83) | -0.03%<br>(-0.18) | 0.37%<br>(1.79)   |                   |
| All Stocks         | -0.01%<br>(-0.33) | 0.16%<br>(2.36)   | 0.53%<br>(5.16) | 0.54%<br>(4.88)  | -0.11%<br>(-1.94) | -0.01%<br>(-0.14) | 0.24%<br>(2.41)   | 0.35%<br>(2.74)   | -0.20%<br>(-2.23) | -0.42%<br>(-4.61) | -0.57%<br>(-4.39) | -0.37%<br>(-2.74) |

**Table 11: Risk-Adjusted Returns of Portfolios sorted on Mispricing , IVOL and TURNVOL**

The table presents the Fama and French five factor alpha of portfolios ranked on mispricing, IVOL and TURNVOL. At the beginning of each month, stocks are sorted into three groups based on their mispricing scores as of the previous month. Within each mispricing groups, the stocks are in turn sorted into terciles on the idiosyncratic volatility (IVOL) as of the previous month. Within each mispricing and IVOL groups, the stocks are in turn sorted into terciles on the standard deviation of *monthly turnover* as of the previous month (TURNVOL). Monthly Turnover is defined as the ratio of monthly trading volume and total shares outstanding in a stock. TURNVOL is computed from 60 months of prior monthly turnover data. In the table TURNVOL is reported as TVOL. Sample period is from January 1966 to December 2013. Returns are value weighted. All t-statistics in parenthesis are computed using the heteroscedasticity-consistent standard errors of White (1980).

|                    | Most Underpriced  |                   |                 |                   |                   | Most Overpriced   |                   |                   |                   |                   |                   |                   |
|--------------------|-------------------|-------------------|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                    | Low<br>TVOL       | 2                 | High<br>TVOL    | High -Low<br>TVOL | Low<br>TVOL       | 2                 | High<br>TVOL      | High -Low<br>TVOL | Low<br>TVOL       | 2                 | High<br>TVOL      | High -Low<br>TVOL |
| Low IVOL           | -0.10%<br>(-1.38) | -0.06%<br>(-0.74) | 0.34%<br>(3.54) | 0.44%<br>(3.53)   | -0.02%<br>(-0.24) | -0.13%<br>(-1.51) | -0.06%<br>(-0.56) | -0.04%<br>(-0.24) | -0.15%<br>(-1.31) | -0.22%<br>(-2.00) | -0.63%<br>(-5.60) | -0.48%<br>(-3.15) |
| 2                  | -0.08%<br>(-0.89) | 0.24%<br>(2.37)   | 0.40%<br>(3.11) | 0.48%<br>(3.05)   | -0.41%<br>(-3.16) | -0.04%<br>(-0.45) | 0.35%<br>(2.90)   | 0.76%<br>(3.98)   | -0.14%<br>(-0.93) | -0.32%<br>(-2.40) | -0.53%<br>(-3.60) | -0.39%<br>(-2.18) |
| High IVOL          | 0.12%<br>(1.03)   | 0.35%<br>(2.62)   | 0.48%<br>(2.63) | 0.36%<br>(1.82)   | -0.05%<br>(-0.32) | -0.05%<br>(-0.34) | -0.07%<br>(-0.40) | -0.02%<br>(-0.09) | -0.55%<br>(-3.29) | -0.90%<br>(-5.84) | -1.13%<br>(-5.49) | -0.58%<br>(-2.40) |
| High - Low<br>IVOL | 0.22%<br>(1.48)   | 0.41%<br>(2.60)   | 0.14%<br>(0.71) |                   | -0.02%<br>(-0.13) | 0.08%<br>(0.46)   | -0.01%<br>(-0.04) |                   | -0.40%<br>(-2.06) | -0.69%<br>(-3.72) | -0.50%<br>(-2.40) |                   |
| All Stocks         | -0.07%<br>(-1.25) | 0.09%<br>(1.42)   | 0.43%<br>(4.43) | 0.49%<br>(4.24)   | -0.13%<br>(-1.72) | -0.06%<br>(-0.93) | 0.16%<br>(1.95)   | 0.29%<br>(2.25)   | -0.16%<br>(-1.57) | -0.29%<br>(-3.32) | -0.57%<br>(-5.28) | -0.41%<br>(-2.92) |